10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 7.2 — Oblique Triangles

Unit 7 | Solved Exercise | New Book (PECTAA)

Exercise 7.2 (Solved)

Law of sines: a ÷ sin α = b ÷ sin β = c ÷ sin γ. Law of cosines: $a^2 = b^2 + c^2 - 2bc \cos \alpha$. Area = ½ ab sin γ (two sides and the included angle) or Heron's formula when the three sides are given.

1. Solve the triangle ABC, in which (all lengths are in cm):

(i) a = 6.1, b = 4.4, α = 42°

(ii) a = 12.2, c = 15.8, γ = 50°

(iii) b = 5.2, c = 5, γ = 48°

(iv) b = 4.8, a = 4, β = 71°

(v) β = 70°, b = 8, α = 100°

(vi) a = 6, α = 55°, γ = 60°

(vii) c = 7, β = 34°, γ = 64°

(viii) b = 12, α = 92°, β = 77°

(i) a = 6.1 cm, b = 4.4 cm, α = 42°

By the law of sines: b / sin β = a / sin α

sin β = (b sin α) / a

= (4.4 × 0.6691) / 6.1 = 0.4827

β = 28.86° ≈ 28.9°

γ = 180° − 42° − 28.86° = 109.14°

c = (a sin γ) / sin α

= (6.1 × 0.9449) / 0.6691 = 8.61 cm

Answer: β ≈ 28.9°, γ ≈ 109.1°, c ≈ 8.61 cm

(ii) a = 12.2 cm, c = 15.8 cm, γ = 50°

sin α = (a sin γ) / c

= (12.2 × 0.7660) / 15.8 = 0.5915

α = 36.26° ≈ 36.3°

β = 180° − 50° − 36.26° = 93.74°

b = (c sin β) / sin γ

= (15.8 × 0.9979) / 0.7660 = 20.58 cm

Answer: α ≈ 36.3°, β ≈ 93.7°, b ≈ 20.58 cm

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(iii) b = 5.2 cm, c = 5 cm, γ = 48°

sin β = (b sin γ) / c

= (5.2 × 0.7431) / 5 = 0.7728

β = 50.61° ≈ 50.6°

α = 180° − 48° − 50.61° = 81.39°

a = (c sin α) / sin γ

= (5 × 0.9887) / 0.7431 = 6.65 cm

Answer: β ≈ 50.6°, α ≈ 81.4°, a ≈ 6.65 cm

(iv) b = 4.8 cm, a = 4 cm, β = 71°

sin α = (a sin β) / b

= (4 × 0.9455) / 4.8 = 0.7879

α = 51.99° ≈ 52.0°

γ = 180° − 71° − 51.99° = 57.01°

c = (b sin γ) / sin β

= (4.8 × 0.8387) / 0.9455 = 4.26 cm

Answer: α ≈ 52.0°, γ ≈ 57.0°, c ≈ 4.26 cm

(v) β = 70°, b = 8 cm, α = 100°

γ = 180° − 100° − 70° = 10°

a = (b sin α) / sin β

= (8 × 0.9848) / 0.9397 = 8.38 cm

c = (b sin γ) / sin β

= (8 × 0.1736) / 0.9397 = 1.48 cm

Answer: γ = 10°, a ≈ 8.38 cm, c ≈ 1.48 cm

(vi) a = 6 cm, α = 55°, γ = 60°

β = 180° − 55° − 60° = 65°

b = (a sin β) / sin α

= (6 × 0.9063) / 0.8192 = 6.64 cm

c = (a sin γ) / sin α

= (6 × 0.8660) / 0.8192 = 6.34 cm

Answer: β = 65°, b ≈ 6.64 cm, c ≈ 6.34 cm

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(vii) c = 7 cm, β = 34°, γ = 64°

α = 180° − 34° − 64° = 82°

a = (c sin α) / sin γ

= (7 × 0.9903) / 0.8988 = 7.71 cm

b = (c sin β) / sin γ

= (7 × 0.5592) / 0.8988 = 4.36 cm

Answer: α = 82°, a ≈ 7.71 cm, b ≈ 4.36 cm

(viii) b = 12 cm, α = 92°, β = 77°

γ = 180° − 92° − 77° = 11°

a = (b sin α) / sin β

= (12 × 0.9994) / 0.9744 = 12.31 cm

c = (b sin γ) / sin β

= (12 × 0.1908) / 0.9744 = 2.35 cm

Answer: γ = 11°, a ≈ 12.31 cm, c ≈ 2.35 cm

2. Calculate the area of each triangle ABC (all lengths are in cm).

(i) a = 7, b = 8, γ = 38°

(ii) a = 11, c = 14, β = 51°

(iii) b = 3, c = 9, α = 78°

(iv) a = 10, α = 62°, β = 69°

(v) c = 4, β = 36°, γ = 80°

(vi) c = 6.6, α = 23°, γ = 89°

(vii) a = 5.3, b = 4.7, c = 8.2

(viii) a = 6.12, b = 8.34, c = 7.12

(i) a = 7 cm, b = 8 cm, γ = 38°

Area = $\frac{1}{2}$ ab sin γ

= $\frac{1}{2}$ × 7 × 8 × sin 38°

= 28 × 0.6157

Answer: Area ≈ 17.24 cm²

(ii) a = 11 cm, c = 14 cm, β = 51°

Area = $\frac{1}{2}$ ac sin β

= $\frac{1}{2}$ × 11 × 14 × sin 51°

= 77 × 0.7771

Answer: Area ≈ 59.84 cm²

(iii) b = 3 cm, c = 9 cm, α = 78°

Area = $\frac{1}{2}$ bc sin α

= $\frac{1}{2}$ × 3 × 9 × sin 78°

= 13.5 × 0.9781

Answer: Area ≈ 13.21 cm²

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(iv) a = 10 cm, α = 62°, β = 69°

First find γ and side b

γ = 180° − 62° − 69° = 49°

b = (a sin β) / sin α = (10 × 0.9336) / 0.8829 = 10.57 cm

Area = $\frac{1}{2}$ ab sin γ

= $\frac{1}{2}$ × 10 × 10.57 × sin 49°

= 52.85 × 0.7547

Answer: Area ≈ 39.90 cm²

(v) c = 4 cm, β = 36°, γ = 80°

α = 180° − 36° − 80° = 64°

a = (c sin α) / sin γ = (4 × 0.8988) / 0.9848 = 3.65 cm

Area = $\frac{1}{2}$ ac sin β

= $\frac{1}{2}$ × 3.65 × 4 × sin 36°

= 7.30 × 0.5878

Answer: Area ≈ 4.29 cm²

(vi) c = 6.6 cm, α = 23°, γ = 89°

β = 180° − 23° − 89° = 68°

a = (c sin α) / sin γ = (6.6 × 0.3907) / 0.9998 = 2.58 cm

Area = $\frac{1}{2}$ ac sin β

= $\frac{1}{2}$ × 2.58 × 6.6 × sin 68°

= 8.51 × 0.9272

Answer: Area ≈ 7.89 cm²

(vii) a = 5.3 cm, b = 4.7 cm, c = 8.2 cm

Three sides are given, so use Heron's formula

s = (a + b + c) / 2 = (5.3 + 4.7 + 8.2) / 2 = 9.1

Area = √(s(s − a)(s − b)(s − c))

= √(9.1 × 3.8 × 4.4 × 0.9)

= √136.94

Answer: Area ≈ 11.70 cm²

(viii) a = 6.12 cm, b = 8.34 cm, c = 7.12 cm

s = (6.12 + 8.34 + 7.12) / 2 = 10.79

Area = √(s(s − a)(s − b)(s − c))

= √(10.79 × 4.67 × 2.45 × 3.67)

= √453.07

Answer: Area ≈ 21.29 cm²

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